

Diffusion Models for Combinatorial Optimization

Undergraduate Thesis

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Industrial Engineering

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2026

Abstract

Keywords: combinatorial optimization, traveling salesman problem, capacitated vehicle routing problem, diffusion models, reinforcement learning, neural combinatorial optimization.

Acknowledgments

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Chapter 1

Introduction

1.1 Motivation

1.2 Problem Statement

1.3 Contributions

- Reproduce and extend *DifusCO*-style diffusion solvers for TSP and CVRP.
- Implement *CADO*-style hybrid fine-tuning and REINFORCE on top of supervised checkpoints.
- Empirical comparison: optimality gap, wall-clock cost, and ablations (reward mode, entropy, denoising steps).

1.4 Thesis Outline

Chapter 2

Background

2.1 Combinatorial Optimization

2.2 Classical and Learning-Based Solvers

2.3 Diffusion Models for COP

2.4 Reinforcement Learning Fine-Tuning

2.5 Related Work

Chapter 3

Methodology

3.1 Overview

3.2 Model Architecture

3.3 Supervised Training

3.4 CADO Fine-Tuning

3.5 Decoding and Evaluation

3.6 Implementation

Chapter 4

Experiments

- 4.1 Experimental Setup
- 4.2 Datasets
- 4.3 Baselines and Metrics
- 4.4 Main Results
- 4.5 Ablations
- 4.6 Discussion

Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

Appendix A

Implementation Details

A.1 Configuration Reference

A.2 Hyperparameter Tables

A.3 Additional Plots

Bibliography